ESSPL
Electronic Safety & Security (Pvt.) Ltd

PROPOSED BY

Electronic Safety & Security (Pvt.) Ltd.

SUBMITTED TO

Ministry of Health, Government of Pakistan

National AIDS Control Programme (NACP) / 1166 Helpline

HIV ANONYMOUS SERVICES PLATFORM

Digital Transformation of 1166 HIV Helpline Services

A Dual-Solution Proposal for Anonymous Intake, Geo-Tagged Case Tracking, and Nationwide Cluster Surveillance

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1. Executive Summary

Pakistan's National AIDS Control Programme operates the 1166 helpline as a primary entry point for citizens seeking HIV information and testing referrals. Today, this service lacks a digital backbone: calls are handled manually, citizens are not registered or tracked, referrals to the 100+ testing centres across the country are verbal, test results are not fed back through the system, and there is no mechanism to identify geographic clusters or outbreak patterns.

ESSPL proposes a technology platform that transforms the 1166 helpline into a structured, anonymous, data-driven HIV services gateway. The proposal presents two solutions, which may be deployed independently or in combination:

- **Solution 1**: A portal-based HIV Unit embedded within the existing 1166 infrastructure, requiring no smartphone or internet access from the citizen.
- **Solution 2**: A dedicated anonymous mobile application for higher-literacy users, future-proofed for expansion to other health verticals.

Both solutions share a common core: strict anonymity for every citizen, a unique Medical Record number (MR#) replacing personal identity, automated referral to the nearest testing center, result delivery back to the citizen, and geo-tagged data feeding a real-time heatmap for outbreak and cluster surveillance by the Ministry of Health.

2. The Challenge

HIV surveillance and treatment uptake in Pakistan faces a set of compounding challenges that a technology platform can directly address:

Challenge	Impact
Stigma and fear of exposure	Citizens avoid testing entirely; high-risk individuals do not call or visit centres due to fear of identity disclosure
No structured intake or tracking	Callers receive a verbal referral but are never registered; there is no follow-up mechanism, no record, and no outcome data
Manual result delivery	Test results rely on the citizen calling back or returning; most do not, leaving cases untreated and untracked
No geographic intelligence	Ministry cannot identify where clusters or outbreaks are forming; resource allocation is not evidence-based
Low digital literacy	A significant proportion of the target demographic cannot use apps or digital portals independently; voice-based services are essential
100+ fragmented testing centres	Centres operate without a unified intake or result system; the same citizen may test at multiple centres without any record linkage

3. Our Approach: Two Solutions

ESSPL proposes two complementary solutions that address different segments of the target population while sharing a common data infrastructure. Both feed into the same central database, the same geo-tagged heatmap, and the same Ministry dashboard.

Solution 1: Helpline Portal	Solution 2: Anonymous Mobile App
For the majority - no smartphone required. Citizens call 1166, speak to the HIV Unit, get registered, and receive an MR# via SMS. Works for low-literacy, feature phone, and no-internet demographics.	For higher-literacy users. Citizens self-register anonymously on the app, get assigned to a testing center, and receive results digitally. No human agent interaction required for sensitive disclosure.

4. Solution 1: HIV Unit within 1166 Helpline (Portal-Based)

This solution integrates a dedicated HIV Unit into the existing 1166 helpline infrastructure. No change is required to the citizen's experience of calling 1166. When a caller's need is identified as HIV-related, they are transferred to or handled within the HIV Unit. From that point, the platform manages registration, referral, follow-up, and result delivery entirely.

4.1 User Flow

Step	Description
1. Call 1166	Citizen calls the 1166 helpline. The HIV Unit option is presented via IVR or transferred by the 1166 agent. The HIV Unit is a ring-fenced service within the helpline, not publicly labelled as HIV-specific to protect caller privacy.
2. Registration	The HIV Unit agent registers the caller in the portal. A unique Medical Record number (MR#) is generated automatically. The MR# is linked to the caller's phone number or CNIC if the citizen chooses to provide it, but neither is displayed on the portal screen - only the MR# is shown to the agent and to the medical center.
3. Location and referral	The citizen provides their nearest locality or landmark (no precise address required). The system identifies the nearest testing center from the 100+ centres database and generates a referral. An SMS is sent to the citizen containing the MR# and the assigned center’s name and address.
4. Test at the center	The citizen presents at the testing center and shows their MR# SMS or verbally provides the number. The center looks up the MR# in the portal and records the test without requiring the citizen's name, CNIC, or any personal identifier.
5. Result entry	The testing center enters the result against the MR# in the portal. The result is flagged to the HIV Unit.
6. Result delivery	The HIV Unit contacts the citizen via the registered number (or sends an SMS where the citizen is not reachable) to communicate the result. All communication is referenced by MR# only.
7. Repeat caller recognition	When a citizen calls again, the phone number or CNIC (if previously provided) flags the existing MR# and full history for the HIV Unit agent, enabling continuity of care without re-registration.
8. Escalation and follow-up	For positive results, the HIV Unit follows a defined follow-up protocol, linking the citizen to treatment and support services by MR# with no name disclosure required.

4.2 System Components

- HIV Unit web portal: secure, role-based access for HIV Unit agents, medical Center staff, and Ministry supervisors.

- Automated MR# generation: unique, non-guessable alphanumeric identifier assigned at registration.
- Testing center database: all 100+ centres mapped with location coordinates and capacity data, enabling nearest-center logic.
- SMS gateway integration: automated SMS to citizen with MR#, center details, and result notifications.
- IVR integration: optional integration with the existing 1166 IVR to present the HIV option and route callers without agent involvement.
- Result entry module for testing centres: simple, secure form for centres to record results against an MR# without storing personal data.
- Follow-up queue: automated queue for the HIV Unit showing all cases with pending results or pending follow-up actions.
- Ministry dashboard and geo-tagged heatmap: all registrations, referrals, and results are anonymously geo-tagged by locality and visualized on a real-time national heatmap for cluster and outbreak surveillance.

4.3 Anonymity and Data Protection

- No name, CNIC, or address is ever displayed to the medical center or to Ministry-level users. The MR# is the sole identifier at all operational touchpoints.
- Phone number and CNIC (where provided) are stored only in a separate, encrypted identity table accessible exclusively to authorized HIV Unit agents for follow-up. This table is never exposed to the portal's general interface.
- Medical centres cannot identify the citizen from any information in the system.
- Ministry-level access is limited to aggregated, anonymized geographic and statistical data.
- All data is stored on Pakistan-based servers. No citizen data is transmitted internationally.

4.4 Geo-Tagged Heatmap and Cluster Surveillance

- Every registration is geo-tagged at locality level (citizen-provided, not GPS-tracked) at the time of call.
- Every confirmed positive result is geo-tagged and contributes to the cluster map.
- The Ministry dashboard displays a live, filterable heatmap by city, district, and province, showing registration density and positivity rates.
- Cluster alerts are triggered automatically when a defined threshold of cases is registered within a defined geographic radius within a defined time window, notifying the Ministry supervisor.
- All heatmap data is fully anonymized. No individual can be identified from any map layer.
- Reports are exportable in PDF and Excel for programme management and donor reporting.

4.5 Advantages of Solution 1

- No smartphone, internet, or app required from the citizen.
- Serves the lowest-literacy, highest-risk demographic that current programmes cannot reach digitally.
- Uses the existing 1166 infrastructure - no new channel to promote or explain to the public.
- Medical centres require minimal training; they only enter an MR# and a result.

- Fastest to deploy and easiest to adopt across 100+ centres nationwide.
- Immediately generates geo-intelligence for Ministry surveillance from day one.

5. Solution 2: HIV Anonymous Mobile Application

This solution provides a standalone mobile application for citizens who prefer to self-manage their HIV testing journey without speaking to a human agent. The app is designed for complete anonymity: no real name, no email address, and no CNIC is required at any stage. Citizens create an account using a self-selected anonymous username and a PIN, making the app safe to use even on a shared device.

5.1 User Flow

Step	Description
1. Anonymous registration	Citizen downloads the app and registers using a self-selected username and PIN. No real identity is required. The system generates an MR# and links it to the username.
2. Location input	Citizen enters their nearest locality or area. The app displays the nearest testing center and provides directions via Google Maps, without tracking the citizen's live location.
3. Booking (optional)	The citizen can request a preferred date and time at the center. The center receives a notification with the MR# and time slot only - no citizen identity.
4. Test at the centre	Citizen presents at the centre and provides their MR#. The centre enters the result against the MR# via the portal.
5. Result notification	The app delivers the result as a push notification to the citizen. The result is viewable only within the app after PIN authentication.
6. History and repeat testing	All previous tests, referrals, and results are stored within the app by MR#, giving the citizen a private, persistent health record.
7. Support and information	The app includes an information module with HIV prevention, treatment, and support resources in Urdu and English.

5.2 System Components

- Flutter mobile application: Android and iOS, offline-capable, with PIN-protected anonymous login.
- Anonymous registration engine: username and PIN only; no email, phone number, or CNIC required.
- Testing centre locator: nearest centre logic using locality input with Google Maps integration for directions.
- Appointment scheduling module: optional slot booking at participating centres.
- Push notification system: result delivery via Firebase Cloud Messaging, fully encrypted.
- In-app result viewer: PIN-gated results page with full test history by MR#.
- Information module: Urdu and English HIV information, prevention guidance, and support service directory.
- Backend API and admin portal: same shared infrastructure as Solution 1, feeding the same national heatmap and Ministry dashboard.

5.3 Future Vertical Expansion

The app is architected as a multi-programme health platform from day one. Future verticals that can be added without rebuilding the system include:

- Tuberculosis (TB) anonymous testing and follow-up.
- Hepatitis B and C screening programmes.
- Mental health referral and support services.
- Maternal and child health programme intake.
- Any NACP or Ministry of Health vertical requiring anonymous intake and geo-tagged surveillance.

Each vertical operates within the same MR# framework, the same anonymity architecture, and feeds the same Ministry surveillance dashboard. The Ministry gains a single platform for multiple sensitive health programmes.

5.4 Advantages of Solution 2

- No agent interaction required, which is critical for sensitive disclosures that citizens may be reluctant to make verbally.
- Citizens manage their own health journey privately on their own device.
- Push notification result delivery is faster and more private than a phone call.
- Persistent anonymous health record gives the citizen and the programme a longitudinal view.
- Future-proofed for multi-programme expansion without additional platform development.
- Lower per-case operational cost at scale compared to agent-handled calls.

6. Solution Comparison

Dimension	Solution 1: Helpline Portal	Solution 2: Mobile App
Citizen requirement	Any phone, can call 1166	Smartphone with internet access
Literacy requirement	Low - agent-assisted	Moderate - self-service
Anonymity level	Agent-mediated, MR# protected	Fully self-managed, no agent
Result delivery	SMS or callback	Push notification in app
Repeat care	Agent looks up MR# on call	Citizen accesses own history in app
Deployment speed	Faster - uses existing infrastructure	Longer - app build and store approval required
Scale for low-literacy	High	Low
Future expansion	Requires new portal modules	Single platform, plug-in verticals
Geo-heatmap contribution	Yes	Yes

Ministry dashboard	Shared - both feed same dashboard	Shared - both feed same dashboard

7. Recommended Approach

ESSPL recommends a phased deployment of both solutions, prioritised as follows:

Phase	Scope
Phase 1 (Immediate)	Deploy Solution 1 - the Helpline Portal - across all 100+ testing centres. This reaches the widest demographic immediately, generates geo-intelligence from day one, and requires no citizen behaviour change. Estimated timeline: 10 to 12 weeks.
Phase 2 (3 to 6 months after Phase 1)	Deploy Solution 2 - the Mobile Application - for higher-literacy users, creating a dual-channel platform. The app builds on the same backend infrastructure already established in Phase 1, significantly reducing development time and cost.
Phase 3 (Post Phase 2)	Expand the app to additional health verticals as directed by the Ministry, leveraging the existing platform without new infrastructure investment.

Deploying both solutions together from the start is also viable if the Ministry wishes to launch a full platform. ESSPL can adjust the timeline and resource plan accordingly.

8. Technical Architecture

Layer	Technology	Purpose
HIV Unit Web Portal	React.js	Agent interface, result entry, follow-up management
Ministry Dashboard	React.js with mapping library	Heatmap, cluster alerts, programme reporting
Mobile App	Flutter (Android and iOS)	Citizen self-service, anonymous registration, results
Backend API	Node.js with NestJS	Business logic, MR# generation, anonymity engine
Database	PostgreSQL	Encrypted, separated identity and operational tables
SMS Gateway	Pakistan-based provider	MR# delivery, result notifications, follow-up alerts
Push Notifications	Firebase Cloud Messaging	App result delivery and alerts
Mapping and Heatmap	Google Maps API	Geo-tagging, centre locator, cluster visualisation

Authentication	JWT with PIN layer	Role-based portal access and app PIN protection
Hosting	Pakistan-based servers	Full data sovereignty, no international transfer

9. Data Privacy and Compliance

- The system is designed from the ground up for strict anonymity. The MR# is the only identifier used at all operational touchpoints outside the HIV Unit.
- Identity data (phone number or CNIC where provided) is stored in a separate encrypted table with access restricted to authorised HIV Unit supervisors only. No other role has access.
- All geo-tagged data used in the heatmap is aggregated at locality level and contains no individual identifier. The minimum geographic unit displayed is sub-district level to prevent individual identification.
- All data is hosted on Pakistan-based infrastructure. No citizen health data is transmitted, stored, or processed outside Pakistan.
- The system architecture is compatible with Pakistan's Prevention of Electronic Crimes Act (PECA) 2016 and the proposed Personal Data Protection Bill.
- ESSPL will sign a formal Data Processing Agreement with the Ministry of Health prior to go-live, defining data ownership (Ministry), data access controls, breach notification timelines, and data retention and deletion policies.
- All medical centre staff and HIV Unit agents will be trained on data protection protocols before system access is granted.

10. Project Timeline

Indicative timeline for the recommended phased approach:

Phase	Duration	Key Deliverables
Phase 1: Discovery and Design	Weeks 1 to 2	Requirements confirmation, system architecture sign-off, data protection agreement, testing centre database compilation, IVR integration specification
Phase 1: Portal Development	Weeks 3 to 8	HIV Unit portal, MR# engine, testing centre module, SMS gateway, result entry interface, follow-up queue, Ministry dashboard and heatmap
Phase 1: Training and Rollout	Weeks 9 to 12	HIV Unit agent training, testing centre onboarding, Ministry dashboard training, go-live
Phase 2: App Development	Months 4 to 7	Flutter app, anonymous registration, centre locator, push notifications, result viewer, information module, app store submission
Phase 2: App Launch	Month 8	Public launch of mobile app, integration with Phase 1 backend, dual-channel operations
Phase 3: Vertical Expansion	As directed	Additional health verticals added to the platform per Ministry direction

All timelines are subject to timely provision of access, testing centre data, and Ministry approvals. ESSPL will confirm a detailed project plan at kick-off.

11. Commercials

ESSPL will draft commercials once the initial proposal is approved by the Ministry.

12. Assumptions and Dependencies

- The Ministry of Health will provide a complete and up-to-date database of all 100+ testing centres including names, locations, and contact details before system development begins.
- The Ministry will provide access to the existing 1166 IVR system and technical documentation for integration.
- The Ministry will nominate a designated focal point with authority to approve design decisions, provide access, and coordinate centre onboarding.
- The Ministry will facilitate testing centre onboarding and ensure centre staff complete the required training before go-live.
- Pakistan-based hosting infrastructure will be procured by ESSPL and costs passed through to the Ministry, or the Ministry may provide its own infrastructure.
- Any change in the scope of work after project kick-off will be handled through a formal Change Request and may affect timelines and cost.
- App store approval timelines for Android and iOS are subject to Google Play and Apple App Store processes and are outside ESSPL's control.

13. About ESSPL

Electronic Safety & Security (Pvt.) Ltd. (ESSPL) is a Karachi-based technology company specialising in the design, development, and deployment of custom digital platforms for public sector and enterprise clients. ESSPL currently operates and maintains several live systems for Karachi Water & Sewerage Corporation (KW&SC), including:

- A Complaint Management Centre (CMC) CRM platform handling thousands of daily complaints across Karachi.
- A unified CEO Dashboard for real-time complaint, hydrant, and legal operations monitoring.
- Hydrant Management Portals with CMC integration and QR-authenticated distribution.
- A Customer Service Centre (CSC) managed services operation.

ESSPL's experience in building anonymous, role-based, geo-tagged platforms for public sector operations makes us uniquely positioned to deliver the HIV platform described in this proposal. We understand the sensitivity of citizen-facing public health systems and the specific challenges of deploying technology in Pakistan's public sector context.

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